



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : BITS F415/ BITS U415
Course Title : Introduction to MEMS
Instructor-in-Charge : Venkatesh Kadbur Prabhakar Rao

1. Course Description:

Micro electro mechanical systems (MEMS), devices and technologies. Micro-machining and microfabrication techniques. Transduction mechanisms and modeling in different energy domains. Analysis of micromachined capacitive, piezoresistive and thermal sensors/actuators and applications. Computer- aided design for MEMS layout, fabrication and analysis.

2. Scope & Objective of the Course:

The course introduces the basic concepts in MEMS (Micro Electromechanical Systems) with a view to address a class of students from science and engineering disciplines. The discussion on topics like MEMS design, Microfabrication, Microfluidics, Sensors and actuators have been structured in the course plan. The objective of the course is to equip the students from various aspects and with basic knowledge of the area of MEMS.

3. Textbooks:

- TB1.** *Micro and Smart Systems*, GK Ananthasuresh, et. al, Wiley-India, 2010.
TB2. *Microsystem Design*, Stephen D. Senturia, Kluwer Academic Publishers, 2001.

References:

- Ref1.** *Fundamentals of Microfabrication*, Marc Madou, CRC Press, 2002.
Ref2. *Foundation of MEMS*, Chang Liu, Pearson Education Inc., NJ, 2006.
Ref3. *An Introduction to Microelectromechanical Systems Engineering*, Nadim Maluf, Artech House, 2000.

4. Course Plan:

Week No.	Module	Lecture No.	Reference	Learning Outcomes
1	Overview of Microsystems	1–2	TB1	Understand the scope, evolution, and applications of microsystems
2	Working Principle of MEMS	3–5	TB1	Explain basic sensing and actuation mechanisms: electrostatic, thermal, piezoelectric
3	Working Principle of MEMS	6	TB1	Compare magnetic and other actuation principles
	Materials for Microsystems	7–8	TB1	Identify suitable materials and evaluate mechanical/electrical properties
4	Mechanics of Microsystems	9–11	TB1	Analyze axial, bending, and torsional behavior in microstructures
5	Mechanics of Microsystems	12–14	TB1	Apply stress-strain and dynamic modeling in microscale structures
6	Mechanics of Microsystems	15–17	TB1	Study energy dissipation, resonance, and quality factor
7	Mechanics of Microsystems	18–19	TB1	Apply FEM and solve numerical mechanical models in MEMS
8	Scaling Laws	20–21	TB1	Understand the effect of miniaturization on physical laws and system behavior
9	Microfabrication	22–24	R2	Learn lithography, doping, oxidation, and etching techniques
10	Microfabrication	25–28	R2	Understand deposition, bonding, and micromachining methods
11	MEMS Devices	29–31	TB1	Analyze pressure sensors, accelerometers, and gyroscopes
12	MEMS Devices	32–34	TB1	Study RF MEMS, micro-mirrors, biomedical and flow sensors
13	MEMS Devices	35–38	TB1	Understand thermal, optical MEMS and reliability issues
14	Characterization & Packaging	39–42	Notes, TB1	Learn microsystem testing methods and packaging strategies

5. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Quiz/ Assignment	10		TBA	
Project work	25			Project work to be discussed in the class
Mid-Semester Examination	25	90 mins.	08 th Oct 2026	Closed Book
Comprehensive Examination	40	180 mins.	11 th Dec 2026	Open Book and Closed Book

6. Notices concerning the course: :

All notices concerning the course will be displayed on the Google classroom

7. Office Consultation Hour:

Will be announced in the class. Also, feel free to email me to seek a prior appointment.

8. Make-up Policy:

Only for Mid-Sem and End-Sem. Make up will be permitted only in cases with appropriate reasons and necessary proof related to the absence.

Signature
Instructor-in-charge
BITS F415